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Editorial Office
Communications Physics
Nature Portfolio

Dear Editor,

We are pleased to submit our manuscript "*Learned variable-support priors accelerate symbolic regression of physical laws*" for consideration as an Article in *Communications Physics*.

What we report. Symbolic regression (SR) is the method of choice for recovering interpretable physical laws from data, but its cost scales steeply with the number of candidate variables and operators, most of which never appear in the true equation. We introduce DenoisedSR, a learned front-end that reads a small set of observations ($q \approx 100$) and predicts which input columns are relevant. The front-end couples a graph-attention network trained over a physical-law co-occurrence graph with a statistical-feature random forest; its output is a reduced variable/operator set that any conventional SR backend can consume. DenoisedSR is therefore solver-agnostic by design.

Why this matters to a broad physics readership. Across four external benchmarks of physical and symbolic laws—AI-Feynman, Strogatz, Nguyen and the SRSD-Feynman suite of Matsubara et al.—the predictor attains recall ≥ 0.96 and never catastrophically discards a true variable. Supplied to a fixed-budget evolutionary backend (PySR), the prior raises exact-law recovery from $42 \pm 13\%$ to a seed-stable $60 \pm 0\%$, shrinks the candidate-column set by 85%, and speeds time-to-solution by up to 6×. The same qualitative gain reproduces on two structurally different backends: classical genetic programming (gplearn) and the 2026 Nature Computational Science cover algorithm PSE (Ruan et al., parallel GPU enumeration); on PSE we observe a +34 to +67 percentage-point lift in exact recovery. The prior is observation-conditioned and adds only ∼40 ms per task, making it a high-leverage, low-risk preprocessing step rather than a competing search algorithm. Beyond the headline metric, our analysis dissects why the prior works (variable selection from column-target statistics, operator selection from physics-knowledge-graph priors), establishes that the gain widens with distractor density rather than benchmark choice, and documents the natural boundary of the approach (the prior is neutral when distractor density is already low).

Fit for the journal. This work sits squarely at the intersection of data-driven physics and machine-learning methodology—two communities *Communications Physics* serves well. We believe it will be of broad interest to physicists who use or develop symbolic regression, and to the wider community working on scientific machine learning. The contribution is complementary to the dimensional-analysis and physics-aware priors that already strengthen scientific SR: we contribute the upstream step of deciding *which measured quantities enter the law at all*.

Open data and code. All experimental artifacts and trained model weights are publicly released on GitHub (`https://github.com/ChenxiHeCam/DenoisedSR`) and archived on Zenodo (10.5281/zenodo.20584889). The release includes paper-reproducing evaluation scripts, all recorded JSON artifacts that back every figure and table, and the trained graph-attention weights.

Originality. The manuscript reports original work, has not been published elsewhere, and is not under consideration by any other journal.

Suggested reviewers (with non-overlapping expertise across SR methodology, neural-symbolic methods, and physical-law discovery; please substitute or amend as the editor sees fit):

- **Miles Cranmer**, University of Cambridge — creator of PySR; expert in symbolic regression for physics. *[suggest: `mc2473@cam.ac.uk`]*
- **Silviu-Marian Udrescu**, Massachusetts Institute of Technology — first author of AI-Feynman; symbolic regression and physics. *[suggest: `sudrescu@mit.edu`]*
- **Yoshitomo Matsubara**, OMRON SINIC X — author of SRSD-Feynman benchmark; SR for scientific discovery. *[suggest: `yoshitomo.matsubara@sinicx.com`]*
- **Kai Ruan**, Renmin University of China — first author of PSE (Nature Computational Science 2026); SR scalability. *[suggest: via group page]*
- **Pierre-Alexandre Kamienny**, Meta AI — end-to-end neural symbolic regression; complementary perspective on assignment problem.

Opposed reviewers. None.

We hope you will find the work suitable for review.

Sincerely yours,

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